Analysis of the Effects of Housing Permits on Population Growth from 2010-2022: Regression Models

Introduction

Population growth is one of the most important factors reflecting how communities change over time, as it can affect (and be affected by) labor market trends, infrastructure quality, public services, and demand for housing, making it crucial to understand the factors that have the strongest effects on population growth. One of the most well studied variables is the availability of housing, as the availability of housing can not only affect the price of housing (thereby affecting the cost of living) but also the physical capacity for a city to hold more people and hire more people within the area itself. House permits, which are a very strong correlate of housing construction, give a clear indication of changes in housing supply, and can be used to examine the relationship between construction and population growth. Additionally, house permitting is affected by a multitude of factors ranging from the amount of land left for zoning, to the regulatory environment in the area. However, population growth is affected by many socioeconomic, demographic, and labor market factors, making the relationship between permitting and population growth far too complicated for simple statistical modeling.

The goal of this project is to evaluate the relationship between housing permits (a strong correlate of house construction) and population growth across U.S. counties from 2010-2022. Using county-level housing, socioeconomic, demographic, and labor market data, this project applies multiple statistical approaches, and machine learning methods to determine the relationship from different perspectives, including linear relationships, nonlinear prediction methods, growth forecasting, and causal inference. This specific report uses linear regression

techniques to examine the relationship between housing permits and population growth, conditional on a plethora of other variables. It gives interpretable coefficient estimates and helps show the extent to which housing permits and other county-level variables are associated with future changes in population.

Data

In order to determine the effect of housing permits on population growth, there were many covariates that were used for different reasons. These included the total permit rate per 1,000 people (obtained from the total number of permits in the county, divided by the population, multiplied by 1,000), as well as the number of single family permits per 1,000, the number of multifamily permits per 1,000 (specifically 2-4 unit permits), and the number of permits for buildings with at least 5 units (per 1,000). Each permit-rate variable was used as the main independent variable for its own regression, while the total permit rate was used for every other type of model. Additionally, the remaining housing variables included county-level homeownership rates, the growth rate in occupied housing units (which was lagged by a year), rent burden (the percentage of people paying more than 30% of their income in rent), and median house value. The labor market and economic characteristics that were used included median household income and the employment population ratio. The demographic variables that were used were the percent of people in a county who are white, the percent of people in a county who are black, the percent of people born outside the US, the change in the percent of people born outside the US (essentially functioning as net migration), as well as the county's median age. Lastly, I also used population density for the counties, which functions as a ceiling on how many people can live in an area before geographic constraints inhibit housing construction (albeit, for

many counties, that limit hasn't been reached). After finding the correct specifications for all of the variables, I also added two interaction terms, permits multiplied by housing growth, and permits multiplied by median house value.

Data Preparation

With the exception of the permit rate, I gathered all of the data from the American Community Survey's 5 year estimates. I used 5 year estimates instead of 1 year estimates because ACS data becomes very noisy when applied to small counties due to their small sample sizes for data collection. For the permit rate, I gathered information on total permits from the Department of Housing and Urban Development, which has the data from 1980-2022, after which I divided the number of each permit type by the population, and I multiplied them by 1,000 to get the per 1,000 rate. I used data from 2010-2022 for almost all of the variables because much of the ACS data starts in 2010, while the housing permit data set runs up to 2022. I also used 2023 data for population, foreign-born population, and the number of occupied housing units. These were used to create growth rates which were mostly used for lags so that I could use the 2022 data to determine trends. The data for the variables (from ACS data) was collected using API keys, and they were merged using SQL joins (mostly inner joins), with some variable transformations and calculations being performed in SQL using window functions, with other transformations being performed either in Excel or in Python workbooks.

For all of the models, I had to make some decisions about what data to include and what to remove. Firstly, since the COVID-19 pandemic falls within the time frame of this project, there are some years where population growth is significantly lower as a result of a crisis that can't easily be seen in the data. The period when this issue was most apparent was 2020-2021, as

population growth was significantly lowered from 2020 to 2021 as hundreds of thousands died from the virus. While even more people died in 2021, this ended up having less of a deleterious effect on population growth because of the substantial increases in immigration during 2021. Since corresponding population growth was the variable of choice, population growth from 2020 to 2021 was associated with the values for each of the variables in 2020. As a result of the issues that 2020 would present for the data, I made the decision to remove the 2020 data from consideration, as well as its corresponding population growth. Although this made models like forecasting far more difficult, the alternative was worse as a whole year's worth of data would be of very poor quality.

Additionally, since there were some very noticeable outliers which could heavily alter the relationship between an independent and dependent variable, I tended towards removing them. This was firstly done by removing counties with a population below 2,500 as population growth figures are incredibly noisy at such low levels, since even a couple families deciding to move out can have a substantial effect on population growth. Additional outliers which were removed included counties which allowed more than 20 permits per 1,000 (a sample of roughly 30 observations), counties which had population growth rates above 8%, and below -5%, (a sample of roughly 40 observations), and counties which saw their number of available housing units either increase by more than 15% in one year, or decrease by more than 12.5% in one year (a sample of roughly 160 observations). While these removed a meaningful amount of data (roughly 0.6% of the data in the sample), even small amounts of outliers were capable of changing the best relationship between an IV and a DV from a linear relationship to a logarithmic one, which is especially problematic for something like regression modeling, where the wrong relationship being specified can make models perform far worse.

Variables

Besides the house permit rate, which is the main independent variable for the project, the housing variables in question were used to communicate how affordable housing is in some way, or to communicate the availability of housing in full. Median house value, rent burden, and homeownership rate are used for different reasons to tell how affordable housing is in terms of the price of housing, the household level constraints from owning housing, or how many people own homes (because many people who don't own houses would own them if they could). The growth in housing units is a reflection of the housing stock at any given time, and even though it could be highly correlated with permitting frequency, it isn't, even when it's lagged.

The employment population ratio was used because it's a simple reflection of how many people work and is superior to the employment rate in some respects because it can better account for an area having lots of discouraged workers, retirees, or limited employment opportunities. Median household income was used to determine how wealthy the average person is in an area.

When it came to demographic variables, I used the three groups (white people, black people, and foreign born people) for different reasons. There is a well documented history of white people moving out of urban areas and into suburban areas when they become more racially diverse, especially with respect to black people moving into an area, and this pattern has been present as suburbs have become more diverse as well (even in middle class neighborhoods), so a higher share of black people in an area may mean that white people move to a nearby county[1][2][3]. I used the white percentage to convey the level of racial homogeneity which may be a correlate of increased housing values for the same reason that an increased presence of minorities may be a correlate of decreased house values. I used the foreign born share not only as

a variable related to diversity, but also as a way to show a crude form of net migration which is itself a massive component of population growth that may not be captured by the data in any other way, especially relevant because the reason for immigrants coming to a certain county aren't likely to be properly predicted from the data. While immigrant flows are somewhat correlated with local labor market conditions, the nature of a labor market can only partially be inferred from the data in question[4].

Lastly, I used population density to communicate the extent to which a county has already been urbanized, its ability to get more permits approved, and its capacity to become more densely populated. I chose this as a measure instead of an urban vs rural percentage because the definition of an urban area has become more restrictive with time, including through an increase in the minimum population requirement.

Methodology

The purpose of linear regression is to find straight lines that minimize the sum of squared residuals (the difference between the actual value of y, and the predicted value), which helps predict what the value of the dependent variable will be, conditional on the value of an independent variable (assuming all else is held constant)[5]. Linear regression was chosen as one of the approaches because it provides easily interpretable coefficients that show the relationship between one variable and another. Once the variables were properly transformed, they're able to provide easily accessible information about the relationship between each variable, and population growth, not only showing the effect that housing permits have on population growth, but the effects of other variables meaning that there are plenty of major takeaways from the results, like the effects of racial and ethnic diversity, living costs, and local economic conditions.

When performing multivariate linear regressions, there are a few fundamental assumptions that need to be met, regardless of whether you're performing ordinary least squares (OLS) regression or linear regressions using scikit-learn (which also runs on OLS). Those assumptions include the model being correctly specified (where a linear relationship between an independent and dependent variable is not represented as a logarithmic relationship, or vice versa), variation in x (essentially that there should be variance in an independent variable so that you know what the effect of a change in that variable is), random sampling, and the zero conditional mean assumption (where the estimated value of the error term, or the difference between the actual and predicted value is equal to zero on average, regardless of the value of any independent variables).

In order to make the proper models, it's critical to ensure that each variable is correctly specified. This is done by checking for the shape of each variable when plotted against population growth, along with the R^2 and the RMSE (root mean squared error) having a minor role in determining whether a transformation is best. However, if the relationship between a given variable and population growth looks linear, I defaulted to using a linear transformation to avoid the risk of fixating too much on small deviations in R^2 and RMSE. Since population growth isn't particularly skewed, it does not require a logarithmic transformation and therefore, the other variables are measured against a dependent variable which has a linear specification. Ultimately, for most variables, a linear specification was adequate. This includes permits per 1,000 (which applies to all 4 of the permit variables), homeownership rate, the employment population ratio, the change in occupied housing units, the change in the percent of foreign born people in a county, the rent burden, the median house value, and the median age in years, as well as the two interaction terms which used linear specifications for both of their variables. The

interaction terms were included because the effect of permits on population growth can be amplified or mitigated based on house prices or growth rate in available housing units. Logarithmic specifications were used for the foreign born population share, as well as population density. A quadratic specification was used for the percent of people in a county who are white, and a cubic specification was used for the percent of people in a county who are black, while a reciprocal specification was used for the median household income.

A model was made for each permit rate variable, including the total rate, the rate for single family permitting, multifamily permitting, and 5 unit permitting, with all having their own permit variable to determine which one has the tightest correlation with population growth overall (all else being held constant). After the OLS models were made, a linear regression was made using scikit-learn, using the total permit rate, with all of the requisite transformations that were made for the prior regressions, with an 80/20 train test split, where the machine learning model was trained on 80% of the data, with 20% of the data used for testing. For the OLS model that didn't run on machine learning, the model performance was judged solely through R^2, and the coefficients were interpreted by looking at the effect of a one standard deviation increase in the variable on population growth. For the linear regression model using scikit-learn, the model performance was judged through the R^2 for both the training and testing data. Lastly, after the scikit-learn model, a plot was made to compare the actual vs predicted values of population growth.

Results

When it came to evaluating the models, the models with the highest R^2 values were the OLS model that looked at total permits per 1,000 (0.477), and single family permits per 1,000 (0.479), while the multifamily permit rate regression (0.366), and the 5+ unit permit rate regression (0.362) had considerably lower R^2 values. For the scikit-learn model, the R^2 for the training set was 0.470, and the R^2 for the test set was 0.494. The scikit-learn regression model also returned very similar coefficient values for almost every variable, with most being within 10% of the value from the baseline OLS model that didn't use machine learning methods. Only two of the coefficients weren't within 20% of the baseline values, with those being median house value, and the reciprocal of median income, where the machine learning model actually flipped the relationship, meaning that higher income correlated with higher population growth, which was not the case in the baseline model.

With respect to the baseline OLS model, the most important variable in driving the value of population growth was total permit rate, with a one standard deviation increase in permitting rate corresponding to a 0.54 percentage point increase in population growth. Meanwhile, the permit*housing value interaction term had a complex relationship, but it tended to push the effect of permitting down as its values have a negative effect on population growth. Additionally, the growth rate in housing units, homeownership rate, median house value, foreign born population share, median age, and the share of black people in a county all corresponded to changes of at least 0.1% in population growth (assuming a one standard deviation increase), with increases in the median age, and the share of black people in a county leading to decreases in population growth rate. One standard deviation increases for population density, median income, and the permit*growth in housing units corresponded to an increase in population growth rate below

0.1%, but above 0.05% (while the permit*housing unit growth interaction term had some cases where the effect was above 0.05% and below 0.05%). One standard deviation increases in rent burden and the linear term for the percent of black people in a county did not have statistically significant coefficients, while the remaining variables had relatively small effects on population growth. Lastly, even though both interaction terms often did have negative contributions to population growth, when you look at the effects of the three variables in question (permit rate, median house value, and corresponding growth in housing units), a one standard deviation increase in the first two terms, coupled with a negative growth rate in housing units (which are the conditions where both interaction terms have negative contributions), the cumulative effect of the three variables is a 0.569% increase in population growth.

When comparing the effects of increases in the rate of each type of housing permit, the effects were actually very similar in spite of the fact that a multifamily permit leads to the construction of more housing units. The coefficients were 0.1708 for the total permit rate, 0.1921 for the single family permit rate, 0.1964 for the multifamily rate, and 0.1889 for the 5+ unit permit rate, with all of the coefficients being statistically significant and having high t-stats indicating that they have incredibly strong relationships with population growth. As was the case with the gradient boosting and neural network models, the relationship between actual values of population growth and predicted values tended to be roughly equal to the R^2 of the model, and not equal to 1, as shown below, with the figure comparing the best fit line of the data using the linear regression model, to a line illustrating a perfect one to one fit:

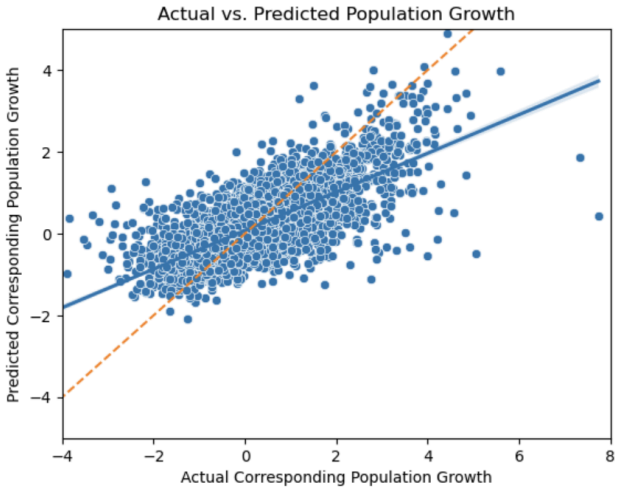


Figure 1: Comparison Between the Prediction For Corresponding Population Growth Rate to the Actual Values of Corresponding Population Growth For Each Observation in the Testing Dataset.

Discussion

The results clearly show a strong positive relationship between house permit rates and subsequent population growth rates at a county level, even after accounting for many other factors. This relationship was very strong, regardless of the major independent variable, be it single unit permits, or 5+ unit permits, while it was also seen in gradient boosting, propensity score matching (causal inference), and forecasting models. The linear regression model made with training and testing data also performed very well, as the coefficients were not only very similar to the values from the baseline OLS regression, but there also was no clear evidence of the machine learning model suffering from overfitting as the R^2 of the test data model was actually slightly higher than that of the training model.

The regression models also found statistically significant associations between almost all variables and population growth, with some variables having larger effects than others. The interaction variables in the regression models were found to be far less important than they were in the gradient boosting models (which said they were generally the most important variables). The effects of each category of permitting were very similar, indicating that no form of permitting didn't have a sizable effect on population growth. This is in spite of the fact that single family permits allow the construction of far fewer housing units than any other permit. This means that based on this model, a single family permit is actually considered far more effective, because a 5 unit permit requires a lot more construction, and space to be taken up, only for the effect of one permit to be almost identical to it. This may be because of differences in the types of counties that permit mostly single unit housing as opposed to those that permit proportionately more multifamily permits.

Additionally, the models for multifamily permits and 5+ unit permits had considerably lower R^2 values, meaning these models were worse at predicting changes in population growth, suggesting that there might be an even more pertinent need to include other variables to make the results a more precise measurement of the effect of multifamily permitting. Those additional variables could ultimately make the multifamily permitting variables have larger impacts on population growth. Lastly, it may simply be a result of the fact that for most counties, the default unit of permitting is single family permits, and that multifamily permits are often not allowed for large areas of zonable land. It's also worth emphasizing that even if each multifamily housing unit isn't as impactful as a single family home, the effect of multifamily permitting is still incredibly large, and there is still merit in building multifamily housing because it's more

affordable, and cost of living issues are a major constraint driving lots of intrastate and interstate migration.

However, these results have a good number of limitations. Firstly, there is a degree of omitted variable bias since variables like crime rate, tax burdens, and public service quality are not included in the model, therefore, the ability to make causal interpretations is somewhat limited. Secondly, there is some reverse causality because population growth incentivizes granting more house permits. Most importantly, the model does not properly match actual values of population growth to predicted values and expected values of population growth tend to be compressed towards zero, as counties with high population growth rates tend to have estimates for population growth that are much lower, while counties experiencing a significant population decline tend to have estimates for population growth that are higher, thereby also closer to zero. This was also an issue with the gradient boosting and neural network models, and it is likely a result of the fact that there isn't enough information to properly predict population growth, since population growth is affected by an array of variables, some of which have data that is relatively easy to collect, while other variables are harder to collect at the county level. This means that an addition to this project could include variables similar to the ones previously mentioned, along with variables for worker protections, transportation quality, and climate.

Overall, the regression models provide evidence of a very strong positive relationship between house permits and population growth. The fact that the relationship is seen across many different permit measures supports the idea that granting permits for housing construction is a very important factor for determining population growth. However, the presence of reverse causality and omitted-variable bias prevent the regression results from giving clear causal interpretations for the effect of permitting on population growth. As a result, the findings suggest

treating permitting as a highly relevant factor when examining population growth, while recognizing that there are many local conditions that need to be considered in order to develop a precise estimate for how impactful housing permits can be.

Appendix

The housing permits dataset was obtained from the department of housing and urban development[6]. The ACS data included the datasets for Selected Social Characteristics, Selected Housing Characteristics, Income in the Past 12 Months, Place of Birth by Poverty Status in the Past 12 Months in the United States, Age and Sex, Hispanic or Latino Origin by Race, Selected Economic Characteristics, and the Annual Geographic Information Table (GEOINFO)[7][8][9][10][11][12][13][14].

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